

MINI PROJECT

SAHAYI– Intelligent Web-Based Micro-Finance & Governance Platform for Self-Help Groups

INTMCA S9

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Abstract

SAHAYI. is an intelligent, secure web-based platform designed to digitize and streamline the financial ledger tracking and governance workflows of neighborhood self-help groups (Ayalkoottams). The platform replaces traditional, error-prone manual paper registers with a centralized digital system that connects group members, group secretaries, and community development administrators to ensure absolute financial transparency and operational accountability.

The platform enables self-help groups to register their units, manage member profiles, schedule weekly meetings, and record real-time attendance. At the core of the system is an automated financial engine that manages the weekly collection of thrift savings and coordinates the entire micro-loan lifecycle. Members can submit digital loan requests directly through the system, which are then routed to the group secretary for review and collective approval. Once approved, the system automatically schedules and tracks repayments, dynamically calculating reducing interest rates. By offering an immutable digital ledger, the platform ensures complete data integrity, prevents financial anomalies, and modernizes grassroots micro-banking.

primary user roles:

1. Group Member

Objective

Allow members to access their financial information and interact with the Self-Help Group digitally.

Features

Profile

- View and update profile
- Change password

Savings

- View savings balance
- View savings transaction history
- Download digital passbook (PDF)
- View savings growth chart

Meetings

- View meeting schedule
- View attendance history
- View meeting minutes

Loan Services

- Apply for a loan
- Upload supporting documents
- Track loan application status
- View repayment schedule
- View loan history
- Download loan statement

Notifications

- Receive meeting reminders
- Receive loan status updates
- Receive savings reminders
- View in-app notifications

2. Group Treasurer

Objective

Manage the financial transactions and loan disbursement of the Self-Help Group

.Features

Savings Management

- Record member savings
- Update savings transactions
- View savings history
- Maintain financial ledger

Loan Management

- Disburse approved loans
- Record EMI payments
- Update outstanding balances
- Generate repayment schedules
- Calculate penalties (if applicable)

Financial Reports

- Monthly financial reports
- Savings reports
- Loan reports
- Export reports (PDF/Excel)

Receipt Management

- Generate digital receipts
- Download receipts
- Maintain transaction history

Personal Features

- View own savings
- Apply for a personal loan
- View personal loan history
- Update profile
- Change password

3. Group Secretary

Objective

Manage the operational activities of an assigned Self-Help Group.

Features

Member Management

- Register new members
- Update member details
- Activate/Deactivate members
- View member profiles

Meeting Management

- Schedule weekly meetings
- Record meeting minutes (MoM)
- Mark attendance
- View attendance history

Savings Management

- Record weekly thrift savings
- Update savings records
- Generate savings receipts

Loan Management

- View loan applications
- Verify member eligibility
- Approve or reject loan applications
- Track loan status

Reports

- Attendance reports
- Meeting reports
- Savings reports

Personal Features

- View own savings
- View own attendance
- Apply for a personal loan
- View personal loan history
- Update profile
- Change password

4. System Administrator (CDS / ADS Level)

Objective

Manage the overall platform, users, and Self-Help Groups while monitoring system activities.

Features

User & Role Management

- Create, edit, and deactivate user accounts
- Assign user roles (Secretary, Treasurer, Member)
- Reset user passwords
- Manage role-based permissions

Self-Help Group Management

- Create and manage Ayalkoottams (SHGs)
- Assign office bearers to each SHG
- View SHG-wise member details

Dashboard & Analytics

- View total SHGs
- View total members
- View total savings
- View active and completed loans
- Financial analytics dashboard
- Attendance and meeting statistics

Reports

- Generate system reports
- Export reports as PDF/Excel
- View financial summaries

Security & System

- Database backup and restore
- Audit logs
- Login history
- System configuration
- Session management

Technologies

- React.js
- ASP.NET Core Web API
- Microsoft SQL Server
- Entity Framework Core
- JWT Authentication
- HTML
- CSS
- JavaScript

Summary

SAHAYI (Phase 1) is a specialized web-based governance and micro-finance platform that eliminates administrative overhead and paper-dependent accounting in neighborhood self-help groups. By utilizing a secure, role-based architecture, the platform automates thrift tracking, attendance logging, and micro-loan processing. This core system establishes a highly transparent, error-free financial ecosystem that empowers local community members, guarantees transaction accountability, and lays a rock-solid data foundation for future mobile and intelligent extensions.